

InternVL3-1B-class and **small Qwen-VL variants** are compact, open-weight **Vision-Language Models (VLMs)** in the **1–4 billion parameter range** 1. Rather than relying on massive, closed, web-hosted models, they represent a class of lightweight multimodal foundation models designed to process both visual inputs and natural language text efficiently 1, 2.

In **SatQuery AI**, one of these 1–4B models serves as the **shared, frozen foundational backbone** across all remote-sensing specialist tasks 1, 3.

1. The Model Families

- **InternVL3-1B-class (e.g., InternVL / RS-InternVL):** An open-weight multimodal vision-language model family developed for high-resolution visual processing and reasoning. It pairs a Vision Transformer (ViT) visual encoder with a lightweight language model decoder, making it particularly effective at visual question answering (VQA) and detailed scene description 1, 4.
- **Small Qwen-VL Variants (e.g., Qwen2-VL 2B–4B):** Open-weight vision-language models developed by Alibaba Cloud. They are widely recognized for strong spatial reasoning, visual token efficiency, and instruction-following capabilities within small parameter footprints 1.

2. Why SatQuery AI Uses a 1–4B VLM Backbone

Fine-Tuned Small VLMs Outperform Larger Generalists

Research across remote-sensing literature demonstrates that a compact 1–4B VLM properly adapted on domain-specific datasets (like **BigEarthNet.txt**) outperforms much larger un-adapted generalist VLMs (such as 8B or 70B models) on specialized geospatial tasks 1, 2.

Avoiding Over-Parameterization & Multi-Task Collapse

Studies on change-VQA highlight a non-monotonic scaling curve: an 8B variant can actually underperform a 4B variant under fixed-rank LoRA fine-tuning on narrow remote-sensing data 1. A 1–4B backbone provides the optimal balance between reasoning capacity and domain adaptability 1.

Parameter-Efficient Adaptation (LoRA)

By using a 1–4B backbone, the heavy vision encoder and base language model remain **completely frozen** 1, 4. Small, modality-specific linear bridge layers and **Low-Rank Adaptation (LoRA)** matrices (rank 8–16) are attached to the attention projections, training **less than 1% of the total parameters** 1, 4.

Inference Efficiency & Hot-Swappable Adapters

- **Single GPU Memory Footprint:** The 1–4B backbone stays loaded in GPU memory once, requiring modest VRAM 1, 5.
- **Zero-Reload Adapter Swapping:** Specialized LoRA deltas for VQA/captioning, grounding, and change analysis are extremely small (**<50 MB each**) and reside in GPU memory simultaneously 1, 6.

- **Low Latency:** When routing compound queries (such as optical-SAR fusion followed by change detection), the serving container hot-swaps active LoRA deltas per forward pass without reloading the base backbone, keeping P95 response latency under **3–8 seconds** 1, 6, 7.

💡 *Would you like to examine how task conditioning tokens (such as [vqa], [ground], or [change]) are prepended to instructions to guide the shared backbone during inference?*